# Gyutae Oh

PH.D. STUDENT · CONTINUAL LEARNING & EFFICIENT LEARNING

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Ph.D. student at the Media System Lab, Sungkyunkwan University (advisor: Prof. Jitae Shin). I build vision models that stay accurate as conditions change — **domain continual learning** without storing past data, **multimodal learning robust to missing inputs**, and **efficient models** with low compute and memory — across medical, industrial, and natural images. 12 peer-reviewed papers (8 first-author) including WACV 2027 and ACCV 2026 (oral); 5 first-author submissions under review (CVPR ×3, ICLR, IEEE JBHI).

## INTERESTS

Domain continual learning · Rehearsal-free / prompt-based CL · Missing-modality learning · Efficient learning & PEFT · Memory-efficient training · Image restoration · Anomaly detection · Medical image analysis

## EDUCATION

**Ph.D. in Electrical and Computer Engineering, Sungkyunkwan University** — Mar 2026 – Feb 2029 (exp.)
Media System Lab · GPA 4.5/4.5 · Thesis (tentative): *Efficient Continual Learning under Distribution Shift*

**M.S. in Electrical and Computer Engineering, Sungkyunkwan University** — Mar 2024 – Feb 2026
Media System Lab · GPA 4.07/4.5 · Thesis: *Efficient Continual Learning Framework for Medical Imaging AI*

**B.S. in Computer Engineering, Korea National University of Transportation** — Mar 2019 – Feb 2024
GPA 3.79/4.5

## EXPERIENCE

**Ph.D. Student (M.S. 2024–2026), Media System Lab, Sungkyunkwan University** — Jan 2024 – Present
Advisor: Prof. Jitae Shin · Continual learning and medical imaging AI

**Undergraduate Researcher, Applied Machine Intelligence Lab (AMI-Lab), KNUT** — Mar 2022 – Jun 2023
Advisor: Prof. Jeonghwan Gwak · Image and biosignal classification

## RESEARCH PROJECTS

**UniPrompt-CL: Single-Inference Domain Continual Learning · `WACV 2027, first author`** — 2025.09 – 2026.10

- Replaced per-layer prompt pools with one pool shared by all ViT layers, so prompt selection and prediction take a single backbone pass; no past data stored.
- Diabetic retinopathy: **+2.0%p** over the strongest same-backbone method, lowest forgetting (AvgF **0.079**), training cost **~¼** of CODA-Prompt (44.17 GFLOPs).

**Multimodal Diagnosis of Exposure Keratopathy & Infectious Keratitis · `with Samsung Medical Center`** — 2024.10 – 2026.09

- Led all stages except clinical data collection (4-person team): design, modeling, experiments, papers, KR/PCT patents.
- Contrastive alignment of four illumination modalities into the broad-beam image at training time; single-image inference reaches accuracy **0.841** / F1 **0.838** (**+16%p** vs. single-modality training).
- Single ViT with mask-aware cross-attention for any 1–4 image combination: 3-class accuracy **0.785 → 0.835**, macro AUROC **0.964** (333 patients, 2,413 sets, 10 repeats).

**NRF Core Research: Adaptive State Inference from Partially Available Multimodal Data** — 2026.09 – (3 yrs)

- ER/ICU deterioration prediction from incomplete ECG, vital-sign, and lab data. One of two core Ph.D. student researchers; managing development by five M.S. researchers.

**Co-authored research (method advisor) · ACCV 2026** 2026.03 – 2026.09

- *Spectral Descriptor Conditioning*: proposed the 2D polar frequency descriptor at the core of the paper — best average PSNR **30.07 dB** over five degradations with only 2.2% extra FLOPs.
- *Drift-adaptive Representation Preservation*: co-designed drift-scaled preservation — **3.0 GB** training memory (4.7–10.7× less) and 13.78 MB storage.

**PatchCore++: Noise-Robust Industrial Anomaly Detection · ICCE-Asia 2024, first author** 2024.04 – 2024.11

- Feature-level noise augmentation with a Locally Aware Common Patch Maker: BTAD instance AUROC **0.9257 → 0.9347**, +2.7%p on the hardest product.

## PUBLICATIONS

12 peer-reviewed papers (8 first-author) · 5 under review · * first / co-first author

### International conferences

1. **UniPrompt-CL: Efficient Single-Inference Continual Learning for Medical Image Classification**
   **Gyutae Oh***, Jitae Shin
   *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2027* · Code
2. **Drift-adaptive Representation Preservation for Domain-Incremental Learning**
   Hyunmin Kong, **Gyutae Oh**, Jitae Shin
   *Asian Conference on Computer Vision (ACCV), 2026*
3. **Change Detection in Longitudinal Chest X-rays Is Readout-Limited, Not Task-Limited**
   Seungin Baek, **Gyutae Oh**, Jitae Shin
   *Asian Conference on Computer Vision (ACCV), 2026* · ORAL
4. **Spectral Descriptor Conditioning: Efficient and Interpretable All-in-One Image Restoration via Polar Frequency Signatures**
   Jungwoo Bae, **Gyutae Oh**, Hyunmin Kong, Jitae Shin
   *Asian Conference on Computer Vision (ACCV), 2026*
5. **Routing Prompts Across Layers: A Global Memory Module for Prompt-Based Continual Learning**
   **Gyutae Oh***, Jitae Shin
   *International Conference on Computer Research and Development (ICCRD), January 24, 2026*
6. **Multimodal Fusion Framework Using Contrastive Learning for Exposure Keratopathy**
   **G Oh***, Y Won, D Lim, J Shin
   *International Workshop on Ophthalmic Medical Image Analysis (OMIA), MICCAI 2025 Workshop, LNCS 16209, 2025* · Paper · Code
7. **Multi-Head CNN-Attention Prompt Pools: Strengthening Prompt Combination and Contextual Insight for Domain Adaptation**
   **G Oh***, J Shin
   *International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), 2025* · Paper
8. **Patch Core++: Augmenting Coreset for Superior Industrial Anomaly Detection**
   **G Oh***, J Shin
   *IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia), 2024* · Paper
9. **Hyperspectral Image Augmentation Technique for the Classification of Diabetic Retinopathy**
   **G Oh***, J Shin
   *International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), 2024* · Paper

### International journals

1. **Causal Representation-Based Personalized Federated Learning with Causal Graph Consensus for Medical Imaging**
   Wooseok Shin, Zhiqiang Shen, **Gyutae Oh**, Jitae Shin
   *Electronics, 15(10), 1983, May 7, 2026* · SCIE · JCR Q2 · Paper

### Domestic conferences

1. **Feature Ensemble-based Wolff Parkinson White Syndrome classification through ECG**
   **G Oh***, I Kim, B Kim, Y Jeon, J Gwak
   *Proceedings of the Korean Society of Computer Information Conference, 2023* · Paper

2. **Identity-CBAM ResNet 기반 얼굴 감정 식별 모듈**
   **오규태***, 김인기, 김범준, 곽정환
   *한국정보처리학회 학술대회논문집 29 (2), 2022* · Paper

**Under review**

1. **Broad-Beam-Anchored Auxiliary-Optional Vision Transformer for Screening and Differential Diagnosis of Exposure Keratopathy and Infectious Keratitis**
   **Gyutae Oh*** et al.
   *IEEE Journal of Biomedical and Health Informatics, 2026* · Co-first author · SCIE, JCR Q1
2. **Double-blind submissions (first author): CVPR 2027 (×3) · ICLR 2027**
   Domain continual learning and efficient learning — titles withheld during review

## PATENT

**Device and Method for Diagnosing Exposure Keratopathy based on Multimodal Fusion Learning** 2025

**KR** 10-2025-0121342 · filed Aug 28, 2025 / **PCT** PCT/KR2026/011329 · filed Aug 10, 2026

Single-image inference reaches **0.841** accuracy and **0.838** F1 on four-level severity grading — an average gain of more than **+16%p** over a single-modality baseline, reducing acquisition burden and improving clinical applicability.

## HONORS

### Awards

**Hanium: Companion Dog Eye Disease Detection Assistant App — Selected Works Prize** 2023.11
Hanium ICT Mentoring Program, hosted by the Ministry of Science and ICT (MSIT), Korea

**Companion Dog Eye Disease Detection Assistant App — Silver Prize & Graduation Project** 2023.08
Korea National University of Transportation

**Hanium: Automated Serving Robot using Object Detection & SLAM — Bronze Prize** 2020.08
Hanium ICT Mentoring Program, hosted by the Ministry of Science and ICT (MSIT), Korea

**Campus Contest: Automated Serving Robot using Object Detection & SLAM — Grand Prize (President's Award)** 2020.09
Korea National University of Transportation

### Scholarships

**SKKU BK21 Fellowship (Specialized Scholarship)** 2024
Sungkyunkwan University

**University Innovation — Merit & Living Expense Scholarship** 2024
Sungkyunkwan University

**Academic Excellence Scholarship (awarded 5 times)** 2019–2023
Korea National University of Transportation

### Service & certification

**Reviewer** 2026
IEEE International Conference on Image Processing (ICIP)

**Advanced Data Analytics Semi-Professional (ADsP)** 2025
Issued by Korea Data Agency (KDATA) · Sep 2025

## SKILLS

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|---|---|
| Core | Python, PyTorch, torchvision, timm, NumPy; Linux multi-GPU experimentation |
| Libraries | Hugging Face Transformers & PEFT, Ultralytics YOLO, scikit-learn, SciPy, Pandas, Albumentations |
| Efficiency | Prompt / prefix tuning, knowledge distillation, FLOPs and GPU-memory profiling |
| Workflow | Git, TensorBoard, Weights & Biases, LaTeX; multi-seed evaluation with confidence intervals |
| Also used | C/C++, Java, TensorFlow/Keras, ONNX, TensorRT, TFLite |

## MILITARY

**Republic of Korea Army — Sergeant, honorably discharged** 2020 – 2022
TOD operator, Gachilbong GOP, 21st Infantry Division (III Corps)